Do Now · retrieval

Year 13 autumn mock: Paper 2

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What makes a program's output testable?

Q2

What is a precondition?

Q3

Which part of a partition is guaranteed to be in its final position?

Q4

Push 4,9,2 onto a stack; pop once; enqueue the popped value to queue [7,8], then dequeue. Give remaining stack and queue.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Answer original practice, not OCR past paper — This diagnostic samples programming and algorithms, including learning from earlier Year 12 lessons.

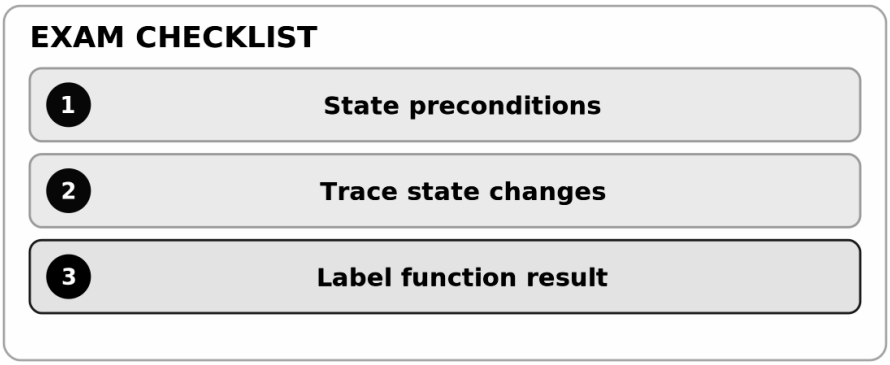
Core idea 2

Time management — Allow roughly 15 minutes per 15 marks.

Core idea 3

Method precision — Binary search must shrink bounds, sorting traces must show complete passes, recursion must include a terminating condition.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

base case — Condition ending recursion.

trace table — Record intermediate algorithm state.

Big O — Asymptotic algorithm growth.

auxiliary space — Extra storage used by an algorithm.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

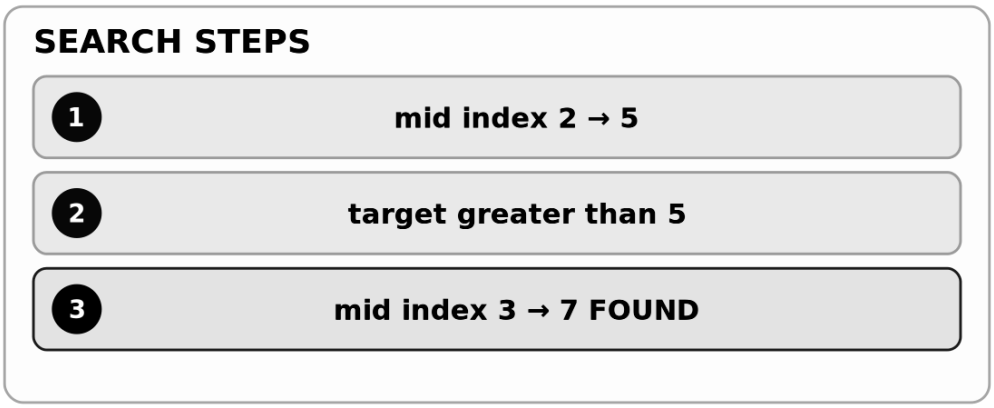
Problem

Unmarked warm-up: trace binary search midpoint for 7 in [1,3,5,7,9].

Follow the worked method

1. Predict the method — Check the middle 5 and discard indexes 0–2 because target 7 is larger.
2. Check and explain — Next middle is index 3 containing 7, so return 3.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the method
Swap 5 and 2 to get [2,5,4].

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

(a) State the key precondition for binary search and explain why it is necessary. [2] (b) Trace binary search for 11 in [1,3,5,7,9,11,13], giving low/high/mid at each step and resulting index. [7] (c) Compare worst-case complexities of linear and binary searches. [6]

Question 2

(a) Starting with [5,1,4,2], show the array after ascending bubble-sort Pass 1, Pass 2 and a final no-swap checking Pass 3. [9] (b) Compare bubble sort with merge sort in worst-case time complexity and auxiliary space. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

(a) Write recursive C# function SumTo(n) returning 1+…+n for n≥0, with base case. [8] (b) Trace SumTo(3) showing the recursive calls and returned value. [5] (c) State one drawback of recursive solutions. [2]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A correctly written recursive function without a base case is guaranteed to terminate.

Use key vocabulary from memory

1. Explain base case in your own words.

2. Explain trace table in your own words.

3. Explain Big O in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.